

Hypothesis: hyperhomocysteinemia is an indicator of oxidant stress.

Elevated plasma homocysteine levels are associated with an increased risk of atherosclerosis and thrombosis, as well as a variety of other pathologies such as birth defects, Alzheimer's disease and other dementias, osteoporosis, diabetes and renal disease. Homocysteine metabolism is catalyzed by a number of enzymes that require B-vitamins as cofactors, and homocysteine levels are particularly responsive to folate status. The predictive power of plasma homocysteine level as a risk factor for atherothrombotic disorders raised the appealing hypothesis that reduction of homocysteine levels by vitamin supplementation might result in a commensurate reduction in the risk of atherothrombotic events. Unfortunately, most clinical trials failed to show a significant benefit of vitamin supplementation on cardiovascular events, in spite of significant lowering of plasma homocysteine levels. Thus, it is not clear whether homocysteine actually plays a causal role in many pathologies with which it is associated, or whether it is instead a marker for some other underlying mechanism. A large body of data links hyperhomocysteinemia and folate status with oxidant stress. In this article I review data that suggests that homocysteine not only promotes cellular and protein injury via oxidant mechanisms, but is also a marker for the presence of pathological oxidant stress. Thus, it is possible that hyperhomocysteinemia is not a common primary cause of atherothrombotic disorders in the general population, but rather a marker of systemic or endothelial oxidant stress that is a major mediator of these disorders. Published by Elsevier Ltd.